

How Do Vectors Add?

SYNOPSIS

Displacement and velocity are vector "counterparts" of distance and speed — they carry a direction as well as a size. Most quantities you'll meet are one or the other, vector or scalar, with no matching partner. Scalars add the way ordinary numbers do. Vectors don't: a vector is an arrow, and arrows add by *joining the head of one to the tail of the next* — the head-to-tail rule. A negative sign flips a vector's head and tail, pointing it the opposite way, which is how vector subtraction works. In this note we learn to add and subtract vectors diagrammatically; multiplication comes later.

PERFORMANCE EXPECTATIONS By the end of this note, you will be able to...

- PE1 Distinguish scalar addition from vector addition** — scalars add like ordinary numbers; vectors add head-to-tail.
- PE2 Apply the head-to-tail rule** to join two or more vectors, keeping each vector's own magnitude and orientation.
- PE3 Recognize that a negative sign flips a vector's head and tail** — it reverses direction without changing length. That's how subtraction works.
- PE4 Draw the resultant vector** — from the tail of the first vector to the head of the last — for two or more vectors joined head-to-tail.
- PE5 Determine a resultant's magnitude and direction by measurement** (ruler and protractor) for vectors joined at any angle.
- PE6 Recognize when vectors must be rejoined** — two vectors drawn joined at the tail (or head), or separated in space, must be moved head-to-tail before you can add them.



You already know this

Head-to-tail addition isn't new — you've been doing it without the vocabulary:



Turn-by-turn directions

"3 blocks east, then 2 blocks north." Each leg starts where the last one ended — that's head-to-tail — and where you end up is the resultant.



Game-controller movement

Press right, then up. Your character's net move is the head-to-tail sum of the two presses, even though you made them one at a time.



Walking it back

Walk 5 blocks east, then 5 blocks west — the second leg is the *negative* of the first. You end up exactly where you started.

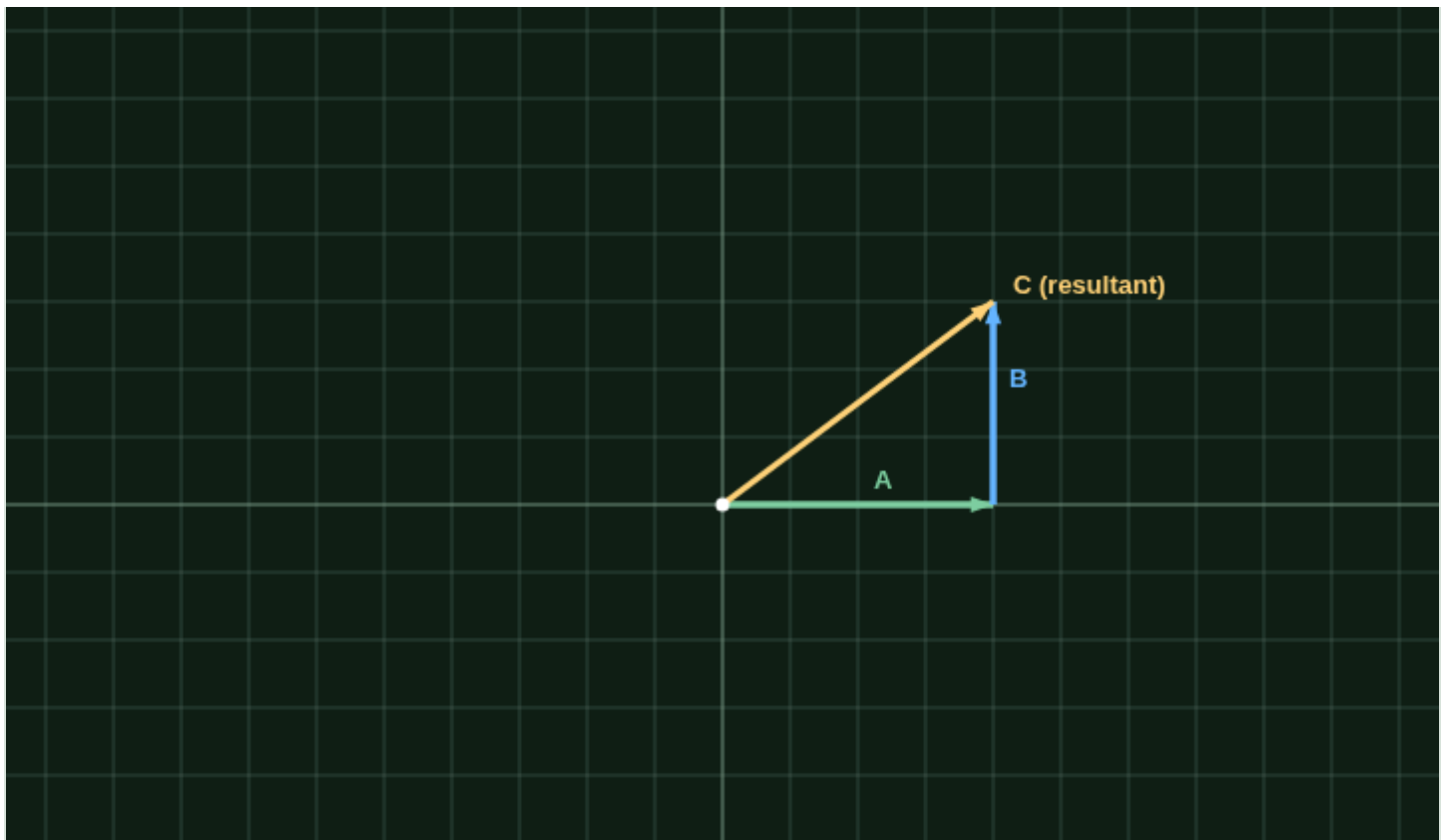
In this note, we work through

- I. What's the difference between a scalar quantity and a vector quantity?
- II. How do vectors add — and why can't we just add their lengths like numbers?
- III. What is the head-to-tail rule, and what does a negative sign do to a vector?
- IV. How do we find a resultant vector's magnitude and direction?

Play with it: head-to-tail vector adder

Set the length and angle of vector **A** and vector **B**. **B** is drawn starting at the head of **A** — head-to-tail — and the resultant **C** runs from the tail of **A** to the head of **B**. Try swapping the order: does the resultant change?

► INTERACTIVE · HEAD-TO-TAIL VECTOR ADDER



Vector A — length

4.0 m

Vector A — angle

0°

Vector B — length

3.0 m

Vector B — angle

90°

↔ Swap order (draw B first)

Show measurement

↺ Reset

Resultant length $\approx$ —

Resultant angle $\approx$ —

Notice: swapping which vector you draw first never changes where the resultant ends up.

Predict first: two vectors, 3 m and 4 m long, are joined head-to-tail at some angle. Which of these magnitudes is *impossible* for the resultant?

7 m

5 m

1 m

8 m

Watch: working through the note

WALKTHROUGH VIDEO · ~10 MIN

Video embed goes here — print the guided notes below before you press play.

Exit ticket

Four quick questions — check yourself before you close this note.

Q1. A vector quantity is best described as one with...

- ☐ Magnitude only.
- ☐ Direction only.
- ☐ Both magnitude and direction.
- ☐ Neither — it's just a number.

Q2. To add two vectors, you must join them...

☐ Tail-to-tail.

☐ Head-to-head.

☐ Head-to-tail — the resultant runs from the tail of the first to the head of the last.

☐ Side by side, without moving either one.

Q3. Two vectors, 5 m and 3 m, point in exactly opposite directions (180° apart) and are joined head-to-tail. What is the magnitude of the resultant?

m

Q4. Two force vectors are drawn joined at their tails. Before you can add them, you must...

☐ Add them exactly as drawn.

☐ Flip one of them.

☐ Separate them and rejoin them head-to-tail, keeping each vector's orientation.

☐ Measure the angle between them first.

Check my answers

Clear

Your notes

BEFORE THE VIDEO

Guided notes — blank (PDF)

Print or open on a tablet and work through the grid-paper vector exercises as you follow along.

